MASHUK ELAHI

Electrical Engineer | Mission-Critical Power Systems & Critical Facilities

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PROFESSIONAL SUMMARY

16 years commissioning mission-critical power systems — MV 11kV switchgear, UPS, BESS, IEC 61850/SCADA protection — in 24/7 zero-downtime environments where a single power excursion scraps a wafer lot; the same N+1 reliability discipline and FAT-to-IST commissioning rigor that semiconductor fabs demand is what Tier III/IV critical facilities require. Delivers the full commissioning lifecycle from Factory Testing (Level 1) through Closeout/Handover (Level 6) across MV/LV power distribution, EPMS/SCADA monitoring, and BESS integration. Certified ISO 9001:2015 & ISO 50001:2018 Lead Auditor (CQI-IRCA) with Python automation experience scaling EPMS load-list and cable-schedule verification.

CORE TECHNICAL COMPETENCIES

- **Power Systems:** MV/LV power distribution · MV 11kV switchgear · transformers · UPS systems (400kVA) · battery / BESS (72kWh Li-ion) · standby generators · chillers / CRAH familiarity
- **Commissioning & Testing:** Integrated Systems Testing (IST) · Factory Acceptance Testing (FAT) · Site Acceptance Test (SAT) · Factory Witness Testing (FWT) · Functional Performance Testing (FPT) · Energisation · systems start-up · protection relay testing · MOP / SOP / EOP · Method Statements
- **Controls & EPMS:** EPMS (Electrical Power Management System) · BMS/BAS · SCADA / PLC (ABB 800xA) · IEC 61850 protection · point-to-point verification · P&IDs · Sequences of Operation · BACnet / Modbus familiarity
- **Standards & Governance:** Tier III / Tier IV design intent · N+1 / 2N+1 redundancy · ISO 9001:2015 & ISO 50001:2018 Lead Auditor · IECEx Ex 001 · CDCP (in progress)
- **Automation:** Python-based EPMS load-list & cable-schedule automation · point-to-point verification at scale · data-driven QA workflows

PROFESSIONAL EXPERIENCE

Electrical Engineer — Mission-Critical Power Systems — Jul 2024 – Present

Strategic Marine Pte. Ltd. · Heavy-Industry Power Systems · Singapore

- Architected the complete LV power distribution system for a multi-generator heavy-industry power plant (3 prime-power generator sets, 450 ekW total) — single-line diagrams, load lists, and cable schedules (Level 2 Installation & Pre-Commissioning) — securing first-pass approval across four independent certification bodies with zero rework cycles.
- Integrated a 72kWh Li-ion Battery Energy Storage System (BESS), BMS, and DC/DC inverters into a hybrid power architecture, cutting grid/fuel energy draw by up to 50% during low-load operation; led BMS-to-EPMS interface integration and Factory Acceptance Testing (FAT) with third-party certification.
- Delivered full electrical design (single-line diagrams, alarm/monitoring I/O lists, cable schedules, IEC-compliant earthing) across a multi-site heavy-industry power programme — dual 380/220V AC + 24V DC distribution, interlocked standby power systems, approved across four independent certification bodies.
- FAT-validated an ABB-based electric-drive Power Management System (EPMS); led site-wide grounding/earthing integrity engineering; standardised departmental deliverables, KPI tracking, and basis-of-design documentation.

Senior Project Engineer — Semiconductor & High-Tech Facilities — Feb 2023 – Jul 2024

Exyte · Semiconductor / Critical Facilities · Singapore

- Led 300mm semiconductor cleanroom critical-power hook-up and Systems Start-up / Energisation (Level 3) to IEC 60204/61511/ISO 14644 — zero audit rework against 100% tool-uptime / N+1 availability targets; PLC/SCADA interlock and Functional Performance Testing (Level 4 FPT) cleared first pass.
- Introduced prefabricated power skids that cut on-site installation labour by ~50% and eliminated contamination risk during live energised operations — a build-out model directly transferable to critical-facility white-space fit-out.
- Drove ISO 50001 energy-efficiency stewardship through smart EPMS load monitoring, measurably reducing facility energy intensity; standardised site Method Statements and MOP/SOP documentation as certified Lead Auditor.

E&I Engineer — Hazardous-Area, Power & Controls — Jan 2021 – Jan 2023

Phoenix Mecano (S.E.A.) Pte Ltd · Industrial Power Systems · Singapore

- Pioneered Malaysia's first solar-wind autonomous remote power installation (24V DC solar/battery bus) — eliminated routine diesel generator use and enabled full remote/unmanned operation, with zero unplanned shutdowns from Energisation (Level 3) through Integrated Systems Testing (Level 5 IST).
- Directed an onshore cogeneration plant electrical & instrumentation scope — transformers, MV/LV switchgear, 400kVA UPS, single-line diagrams, tie-in and instrument-loop design; SCADA (ABB / Siemens / Yokogawa) integration and technical bid evaluation for all major electrical packages.
- Designed IECEx/ATEX Zone 1/2 hazardous-area systems — Ex equipment selection, intrinsically-safe barriers, zoning studies — and standardised compliance documentation as certified ISO 9001 / ISO 50001 Lead Auditor.

E&I Superintendent — Heavy-Industry Power Systems — Nov 2012 – Dec 2020

Keppel FELS · Heavy-Industry Power Systems · Singapore

- Superintended crews of up to 100+ across fabrication, installation, and integrated commissioning (Level 1 FAT through Level 6 Closeout/Handover) of large-scale heavy-industry power systems — energised MV 11kV switchgear and ABB 800xA SCADA-based fire & gas protection; maintained MV/LV power reliability at 100% uptime through final systems turnover.
- Energised a wind-power substation electrical system to 100% IEC 61850 protection-relay conformity — full protection-relay coordination (Level 3 Energisation), alarm/SCADA validation (Level 4 FPT), and final commissioning sign-off (Level 5 IST); zero punch-list items outstanding at handover.
- Commissioned large-scale mobile heavy-industry power and accommodation modules through FMEA, cause-and-effect matrices, Factory Acceptance Testing (FAT), and Site Acceptance Testing (SAT); mentored 15+ engineers via Innovation & Quality Circles.
- Authored 30+ electrical & instrumentation risk assessments; delivered integrated control, alarm, and monitoring instrumentation systems; led EPC detailed-engineering deliverables across multiple concurrent heavy-industry power builds.

Assistant Electrical Engineer — Industrial & Infrastructure Projects **Sep 2009 – Oct 2012**

Ananda Shipyard & Slipways Ltd · Bangladesh / APAC

- Led electrical & instrumentation crews of 15–30 on newbuild heavy-industry power projects — commissioned large rotating-machine drive systems and integrated control equipment to third-party certification and owner acceptance, achieving first-time milestone sign-off.
- Validated cable routing and interconnection designs against load schedules and voltage-drop criteria before energisation (Level 2 Installation & Pre-Commissioning); governed BOM and delivery scheduling to protect on-site readiness against supply-chain disruption.
- Delivered LV/MV power distribution, protection coordination, and commissioning across industrial and infrastructure projects — establishing the foundation of 16 years' uninterrupted career in electrical engineering.

EDUCATION & PROFESSIONAL CREDENTIALS

Education

- MSc Sustainable Electricity Generation — Universidad Católica de Murcia (UCAM), Spain — Apr 2026
- MSc Digital Transformation — London Metropolitan University, UK — Mar 2026
- Master of Business Administration (MBA) — University of Sunderland, UK — Jun 2025
- BSc Electrical & Electronic Engineering — Bangladesh University — Jul 2008
- Diploma in Electronics Engineering — Jessore Polytechnic Institute — Jan 2004

Certifications

- CDCP (Certified Data Centre Professional)
- ISO 50001:2018 Energy Management — Lead Auditor (CQI-IRCA)
- ISO 9001:2015 Quality Management — Lead Auditor (CQI-IRCA)
- IECEx Ex 001 — Basic Principles of Protection in Explosive Atmospheres (valid to Oct 2027)
- ABB — Power System / LV Switchboard Training
- Fluke DSX2-8000 Cable Analyzer Copper Certifier Training